

Optimizing AI-Generated Storyboards for Enhanced Script Alignment

1. Introduction

In recent years, the micro-drama market represented by ReelShort has grown rapidly. Consequently, the demand for content production has significantly increased. However, creators still rely on manual work to complete tasks such as script splitting, shot design, and prompt writing. A short drama of only 2 minutes typically requires 20 to 30 shots. The research topic of this project is an AI short drama storyboard automation system based on multi-agent technology. It aims to address the issues of low efficiency in storyboard production, inconsistent visual styles, and high platform adaptation costs in current AI short drama creation. And manual design is not only time-consuming but also prone to problems, such as inconsistent character images, scene slips, and chaotic lighting and color tones. Additionally, different video generation platforms (Seedance 2.0) have specific input formats and parameter rules. So manual adjustments further increase the creation threshold. As noted by Anantrasirichai and Bull (2022), the integration of AI in the creative industries has shifted from simple tool-based assistance to a collaborative intelligence model that can significantly optimize production pipelines.

2. Problem Analysis & Research Topic

Currently, short drama storyboard creation relies heavily on either manual work or simple conversations with ordinary LLMs (e.g., ChatGPT). Both approaches have significant inefficiencies and drawbacks, which directly result in what we define as “incomplete storyboarding” — a set of recurring symptoms that undermine production quality and efficiency.

Symptoms of Incomplete Storyboarding:

- Vague Shot Sizes: Failure to distinguish between Close-ups (CU), Medium Shots (MS), and Wide Shots (WS).
- Erratic Camera Logic: Inconsistent or illogical movements (e.g., poorly timed pans, tilts, or tracking shots).
- Sequence Fragmentation: Lack of continuity between consecutive shots.
- Tone Mismatch: Visuals that do not align with the script’s emotional arc (e.g., comedic framing for a tragic scene).
- Lack of Detail: Insufficient descriptions regarding composition and character blocking.

Root Causes of These Symptoms:

This phenomenon of "visual inconsistency" remains a critical bottleneck in text-to-video generation, as general-purpose LLMs often lack the spatial reasoning required to maintain semantic continuity across multiple frames (Ramesh et al., 2022).

- **Ordinary LLMs:** lack visual preview capability and suffer from difficult local editing (modifying one shot requires regenerating the entire sequence).
- **Generation platforms (e.g., Seedance):** lack cross-shot consistency (character/scene/color drift) and have no structured control interface.
- **LLM+platform combination:** the LLM cannot enforce visual consistency, and the platform does not automatically align frames across shots; adjusting a single shot forces a full re-execution of the entire pipeline, which is inefficient and not scalable.

Impact: Production delays, increased manual correction costs, and a disconnect between the original creative vision and the final output.

Scope: This project focuses on optimizing the output of AI storyboard generators rather than developing a new base video engine from scratch.

3. Why is Multi-Agent the right tool?

The multi-agent system solves the above problems by directly addressing each root cause. According to the multi-agent conversation framework proposed by Wu et al. (2023), decomposing complex tasks into specialized roles allows for autonomous error correction and a significant reduction in hallucination during the creative reasoning process.

- First, local editability allows users to modify a single shot without breaking narrative coherence, eliminating the ordinary LLM's need for full-sequence regeneration.
- Second, instant visual preview automatically generates frames from shot descriptions, removing the LLM's lack of vision and the manual copy-paste to separate platforms.
- Third, cross-shot consistency reuses fixed character, scene, and lighting descriptors across all shots, solving generation platforms' drift and lack of structured control.

Together, these three capabilities systematically eliminate the symptoms of incomplete storyboarding and reduce production delays, correction costs, and vision-to-output disconnect.

4. Methodology & Implementation Path

4.1 General Approach

This project was developed and run on the Google Colab platform. Based on the large language model API, a multi-agent short drama storyboard automation system was built. The entire process, from script input to generating custom prompts, is automated. The system is written in Python and uses Streamlit to build an interactive interface. Furthermore ngrok is used to enable public network access, facilitating demonstrations and tests. The implementation of the entire

system is divided into four core parts: environment configuration, multi-agent architecture, process scheduling, interface deployment and operation.

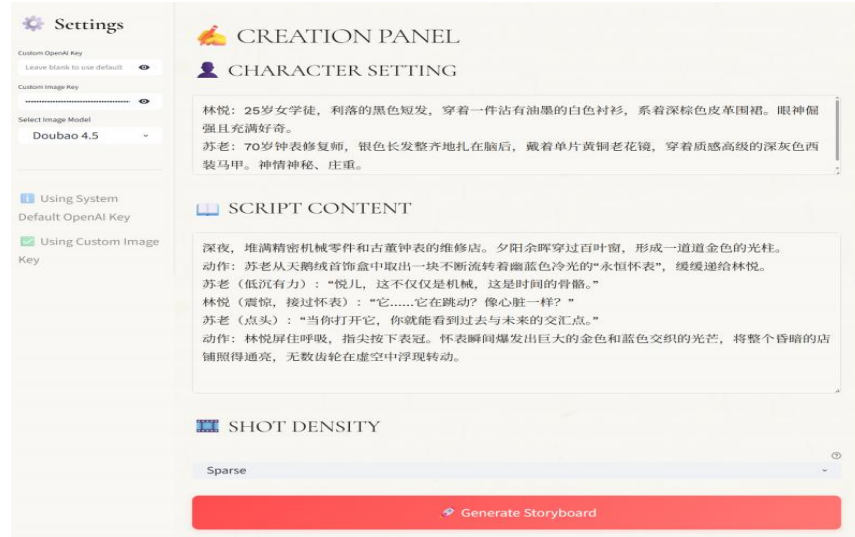


Figure 1: The user interface of the AI Storyboard System

4.2 Implementation Details

- Multi-Agent Architecture:** The core of the system is a three-layer intelligent agent architecture, uniformly managed by the overall orchestrator, with clear divisions of labor:
 - Director Agent (DirectorAgent):** Responsible for script parsing, extracting characters, scenes, emotions, actions and plot rhythm. Output is structured information, and automatically determining the recommended number of shots, ensuring reasonable storyboard length.
 - Cinematography Agent (CinematographyAgent):** This agent executes storyboard disassembly and visual composition. To ensure technical precision, it utilizes constraint-based system prompts that mandate the inclusion of specific camera parameters (e.g., FOV, Lighting Temperature). This forces the LLM to move beyond vague descriptions and adopt professional cinematography language, ensuring consistent character blocking, lighting, and color tones across the entire sequence.

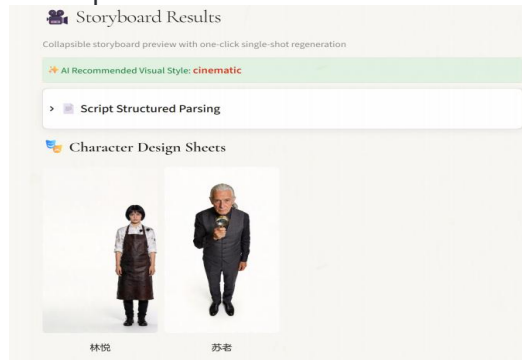


Figure 2: Character Design Sheets

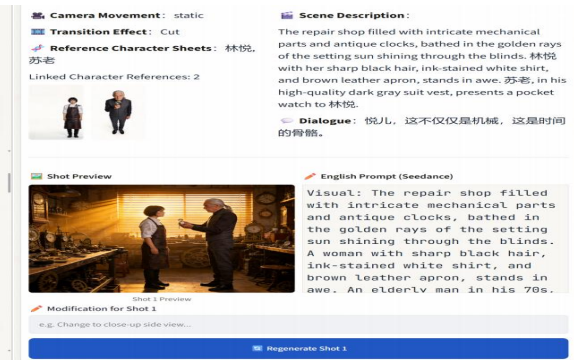


Figure 3: Shot Design

- **TechTransAgent (TechTransAgent):** Converts Chinese shot descriptions into high-quality English prompts that meet Seedance 2.0 requirements. This specific optimization ensures to ensure that the character details, lighting, color tone, and art style were fully preserved.
- **Process Scheduling:** These three agents are uniformly controlled by the overall orchestrator. And executing in sequence: receiving user input → parsing the script → generating storyboard sequence → generating prompts → rendering keyframe images. The system also supports single-shot regeneration functionality, which can self-correct based on the context of previous and subsequent shots to maintain narrative coherence. The system then calls the Seedance 2.0 image generation API with each prompt, retrieves the resulting keyframe images (PNG), and displays them in the interface as instant visual previews.
- **Interface Deployment & Operation:** At the interaction level, the system uses Streamlit to build a visual web interface. Users only need to input three items: short drama script, character setting, and shot density. The system can automatically output character setting diagrams, complete storyboard sequences, keyframe previews. And support the export of storyboard tables in TXT, Word, and Excel formats. After the code is finalized, the Streamlit app launches in the background and bridges to a public ngrok URL. This provides a live, accessible interface for real-time demonstrations and file exports, turning the project into a fully functional, reproducible engineering solution.

5. Results & Conclusion

5.1 Project Outcomes

This project, based on a multi-agent architecture, completed the development and testing of an AI-powered automated storyboard system for short dramas, achieving one-click generation from script input to professional storyboard sequences and Seedance-compatible prompts. Through multiple script tests and Seedance 2.0 video generation verification, the system can stably output structured storyboards conforming to film and television audiovisual grammar. Elements such as shot type, camera movement, duration control, and composition are complete and standardized, maintaining a high degree of consistency in character imagery, scene space, and lighting and color tones. This effectively avoids common problems in traditional AI generation, such as chaotic shot logic, image drift, and disjointed narrative rhythm.

Taking the "Rooftop Evening" scene as an example, the storyboard generated by this system clearly marks shot types such as low-angle shots, close-ups, and long shots. Combined with slow panning and pull-out camera movements, and strictly setting the duration of each shot, a progressive visual rhythm of low angle—close-up—long shot is formed, resulting in clear narrative and professional cinematic language. Compared to unstructured text generated by ordinary LLMs, this system's output exhibits stronger executability and generation stability, directly driving video models to achieve high-quality image rendering without requiring manual secondary modifications. Overall results show that this system significantly outperforms

[illegible]

5.2 Conclusion

Based on the empirical evaluation framework for storyboard quality proposed by Truong et al. (2006), high-quality storyboards must meet five core elements: clear information, coherent narrative, appropriate detail, reasonable number of shots, and clear camera movement logic. Kalimuthu & Kalaiselvan (2025) further emphasized in their research on storyboard art that professional storyboards should possess standardized camera grammar, stable spatial relationships, and a rhythmic design that fits the emotions. The storyboards generated by this system strictly adhere to the above standards – for example, employing a three-shot structure to fit the optimal expression length of a single narrative unit, shot progression conforming to visual reading logic, focusing visual information on core content, and precisely matching camera movement and duration to the emotional atmosphere. In terms of standardization, professionalism, and coherence, they meet the requirements of pre-production film and television creation.

Comparative results demonstrate that while ordinary AI-generated text often fails to maintain cross-shot consistency, our multi-agent system transforms scripts into directly executable storyboards. By aligning with the concept of "Co-Creativity" (Oppenlaender, 2022), the system serves as an extension of human artistic intent, significantly improving narrative clarity and visual stability. Future optimizations will focus on complex action scheduling and real-time pre-visualization, providing a more robust automated solution for the evolving AI film and television

industry.

6. Workload Allocation

A transparent breakdown of team contributions.

- [Chen Zitao]: System Integration & Code Refactoring.
- [Zeng Ziyun]: Multi-Agent Workflow Design & Logic Mapping.
- [Li Junze]: Comparative Quality Assessment & Video Verification.
- [Liu Jiayu]: Prompt Engineering & Narrative Alignment Analysis.
- [Du Yu]: Core Backend Development & API Orchestration.

7. References

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